OpenGeoStreams in plain words

A Python library and web dashboard that turn river water-quality spreadsheets into clean data, charts, safety checks and maps that follow the river.

1. What it is

Monitoring agencies publish river water-quality data as spreadsheets in many layouts: different column names, date styles, station spellings and units. OpenGeoStreams reads any of them and gives you one consistent dataset to work with.

In a notebook or script

Works like pandas: load a file, then ask it questions.

```
import opengeostreams as ogs

data = ogs.load_csv("river.csv", rows="all")
data.describe()      # statistics
data.plot_trend(parameter="BOD")
data.interpolate()   # along the river
```

In the browser

One line opens an interactive map and charts for the same data. You can also drop new files straight into the page.

```
data.dashboard()
# opens http://127.0.0.1:4173
```

Both use the same Python code, so the notebook and the dashboard always agree.

> The project's case study is the N-choe, a seasonal stream through Chandigarh and Mohali that joins the Ghaggar: 1,560 records from 13 stations, 2018 to 2025. Nothing in the library is specific to it; it works for any river.

2. The five layers

Read left to right. Each layer takes the result of the one before it.

1 → 2 → 3 → 4 → 5

1 Bring in data

CSV, Excel, JSON or a pandas table. Any column names, any date style.

`load_csv` · `concat`

2 Clean it

One standard table: parameter names, dates, units and station spellings made consistent.

`pipeline.py`

3 Find the river

Puts each station on its drain or river, works out which way water flows, and orders stations upstream to downstream.

`network.py`

4 Analyse

Trends with rainfall, distributions, fitness for drinking, farming, fish and bathing, and estimates along the river with a self-check.

`plots.py` · `qualitycheck.py` · `interpolation.py`

5 Share

Figures and tables in Python, or the interactive map dashboard in a browser.

`dataset.py` · `server.py` · `dashboard`

Outside data it uses

SOURCE	USED FOR	NEEDED?
India-WRIS river map	Drains and rivers in India (bundled for Punjab)	Only for river-aware features
HydroRIVERS or your own map	Rivers anywhere in the world, or your own drain and canal lines	Optional
Open-Meteo	Rainfall beside each trend, and ground height to decide which way channels flow	Optional (internet)
Google Earth Engine	Station elevations	Optional (account)

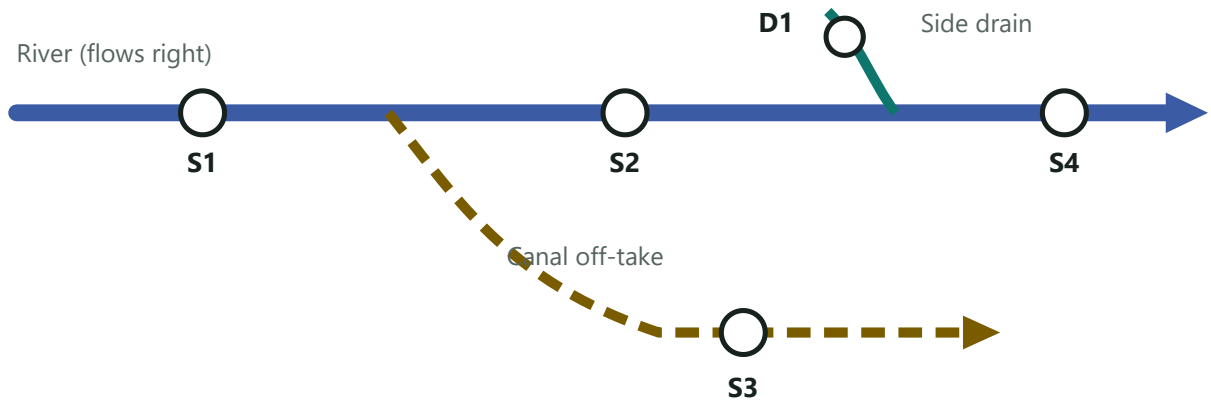
3. What happens to your data

From the moment you load a file to the moment you see a chart.

1. **Read.** The file is read as rows. You choose how many (`rows=400` by default, or `rows="all"`).
2. **Standardise.** Every row becomes: Parameter, Unit, Date, Value, Source, Location, Latitude, Longitude. Aliases such as "T.Coliform" become "Total Coliform". Dates like "Jan 2021" or "2023" become proper date ranges.
3. **Merge station spellings.** Labels such as "POINT SOURCE BUDHA NALLAH" and "SOURCE BUDHA NALLAH 30.973" become one station. Upstream (U/S) and downstream (D/S) stations are never merged.
4. **Check.** Empty cells are counted, and stations whose rows disagree about their coordinates are flagged so you can fix the file.
5. **Analyse on demand.** Charts, river matching and interpolation run only when you ask for them, and their results are kept so they aren't recomputed.

4. Following the river

Water flows along channels, so a station only tells you about places its water actually reaches. OpenGeoStreams measures distance along the river, never in straight lines across land.



- **S1 counts for S3.** The canal takes its water from above the off-take, so S3's water passed S1.
- **S2 and S3 never count for each other.** They are on different branches below the split, so no water flows between them.
- **D1 counts for S4 but not for S2.** The drain only joins the river below S2.
- **At a junction, the bigger branch counts more** when catchment areas are known; otherwise both sides count equally.

How a value is filled in along the river

1. **One interval per station.** An ordinary reading is one value; a Min/Max report is a range. Every year counts equally.
2. **Two runs, one interval.** The map is built once with the Min of every range (the lower bound) and once with the Max (the upper bound), so stations that only report a range still count. A station with only a Max counts for the upper bound only.
3. **Average the connected stations.** Each point on the river averages the stations upstream and downstream of it. Nearer stations, and stations that supplied more of its water, count more.
4. **Let the data choose the reach.** Limits of 2, 4, 8, 16 and 32 km and no limit are tested, and the one that predicts best is kept.
5. **Check itself.** Each station is hidden in turn and predicted from the others. The map reports the typical miss and a *skill* score: above 0 means it beats simply averaging the other stations.
6. **Show honestly.** Colour is drawn only along mapped channels and fades where confidence is low. Without a river map it falls back to straight-line distance and says so.

> With 50 or more stations you can also ask for stream-network kriging (`interpolate(advanced=True)`). It is kept only if it beats the simple method in the self-check. The approach follows Ver Hoef & Peterson's stream-network statistics.

5. The dashboard

A map-first screen. Everything you see is computed by the same Python library.

PART OF THE SCREEN	WHAT IT SHOWS	MADE BY
Left panel	Your files (import, combine, download), parameters and the station list	`dataset.py`, `pipeline.py`
Map: station dots	Each station, coloured pass / caution / fail for the chosen use	`qualitycheck.py`
Map: blue lines	The drains and rivers each station was matched to	`network.py`

Map: coloured ribbon	Estimated values along the river, faded where confidence is low. A Lower / Upper switch picks the bound; click for the full interval	`interpolation.py`
Use-case card	Fitness for drinking, farming, aquatic life or bathing, with reasons	`qualitycheck.py`
Analysis drawer: Trend	The parameter over time, with rainfall bars	`plots.py` + Open-Meteo
Analysis drawer: Distribution	Box plots by year or month	`plots.py`
Analysis drawer: Stream	Station × year grid ordered upstream to downstream, per drain	`plots.py`, `network.py`
Period bar	Filter by year or month	the browser

The browser only draws. Python builds each chart and sends it as data, so a chart looks the same in the dashboard and in a notebook.

6. Using it from Python

Everything starts from one object, a `RiverDataset`.

GROUP	FUNCTION	WHAT IT DOES
Load	`ogs.load_csv(path, rows=400)`	Read a file into a dataset (`rows="all"` for everything)
	`ogs.from_dataframe(frame)`	Make a dataset from a pandas table
	`ogs.concat([a, b])`	Combine datasets or tables
Look	`data.head()`, `data.describe()`	First rows; statistics per parameter
	`data.summary`	Rows, stations, parameters, empty cells, merged names
	`data.coordinate_conflicts()`	Stations whose rows give different positions

Rivers	`data.channel_network()`	Match stations to drains and rivers, upstream to downstream
	`data.assign_channels()`	The same as a table, one row per station
	`data.fetch_elevations()`	Station elevations from Google Earth Engine
Analyse	`data.interpolate()`	Values along the river, with the self-check
Show	`data.plot_trend()`	Parameter over time with rainfall
	`data.plot_distribution()`	Box plots by year or month
	`data.plot_stream()`	Station × year grid, upstream to downstream
	`data.plot()`, `data.plot_map()`	The interpolated map; an interactive web map
Share	`data.to_csv(path)`	Save the cleaned data
	`data.dashboard()`	Open the interactive dashboard

Every `plot_…` function also takes `save_path="figure.png"` or `"figure.html"`. The notebook `examples/opengeostreams_walkthrough.ipynb` runs each one, and every cell runs on its own.

7. Where things live

FILE	IN ONE LINE
`dataset.py`	The `RiverDataset` object you use; hands work to the files below
`pipeline.py`	Reads and cleans files; builds the data the dashboard starts from
`network.py`	Reads river maps; matches stations to drains and rivers
`interpolation.py`	Fills in values along the river and checks itself
`qualitycheck.py`	Pass / caution / fail rules for each water use
`plots.py`	Every chart and the web map
`elevation.py`	Optional Google Earth Engine elevations

`server.py`	The small web server behind the dashboard
`dashboard_assets/`	The dashboard page: map, panels and chart drawing
`reference_data/`	Bundled Punjab drain map and local river-name aliases

Function-by-function details, web endpoints and how requests flow are in the developer reference.

8. Words used here

Station

A place where water samples are taken.

Parameter

What is measured: pH, BOD, nitrate, coliform and so on.

Min / Max readings

The lowest and highest value reported for a period. They give the lower and upper bound of the map's interval.

Choe, nallah, drain

Local names for seasonal streams and drainage channels.

Upstream / downstream

Towards the source / towards where the water ends up. Labels U/S and D/S.

Junction / off-take

Where two channels meet / where a canal takes water out of a river.

IDW

Inverse distance weighting: nearer stations count more in an average.

Leave-one-out

Hide one station, predict it from the rest, repeat for every station.

Skill

How much better the map predicts than simply averaging the other stations. Above 0 is better.

Kriging

A statistical way of filling in values that fits how quickly they change with distance.